Ondřej Baštař

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RESEARCH EXPERIENCE

SPAR (Supervised Program for Alignment Research) *Research Fellow* **Sep 2026 – Present**

- *Coherence-preserving steering via learned activation denoisers*, mentored by Francisco Ferreira da Silva (Pivotal Research). Replicating GLP (Luo et al., ICML 2026), a diffusion model of LLM activations used as an on-manifold steering prior, on Llama-3.2-1B.
- Developing a Cold Diffusion variant that replaces Gaussian noise with deterministic degradation along collected steering directions.

FEE CTU *Researcher, Bayesian Inverse Reinforcement Learning* **2025 – Present**

- With Ondřej Bajgar, developing QVIRL / ValueWalk, a Bayesian method that infers what an agent is trying to achieve, with uncertainty, from demonstrations of its behaviour.
- The method works on small discrete toy problems; scaling it up to high-dimensional continuous control, such as simulated robot locomotion (MuJoCo).

AI Center (FEE CTU) *NLP Research Intern* **2022 – 2023**

- Sentence-to-sentence similarity: built a dataset that mines naturally similar sentence pairs from cross-source news coverage of the same events.
- Experimented with transformer language models of the time (BERT and its variants) for sentence-similarity tasks.

WORK EXPERIENCE

Vrgineers *Mixed Reality Researcher / Developer* **2023 – Present**

- Lead developer on the mixed reality pipeline for the XTAL / Somnium VR1 headsets.
- Built a real-time C++/CUDA image processing pipeline for 5000×5000 @ 102 fps video (demosaicking, denoising, sharpening, distortion correction, …).
- Optimised the pipeline stages with low-level CUDA work, e.g. making the denoiser's cost negligible at unchanged quality.

FEE CTU *Teaching Assistant* **2025 – Present**

- First-year Linear Algebra tutorials for Open Informatics students, a role usually held by mathematics PhD students.

COMTES FHT *Engineer* **2020 – 2022**

- Prototype metal 3D printer, IoT machine monitoring, and an ImGui/Python tool for training image-analysis networks.

Whitesoft *Freelance IoT Engineer* **Summer 2020**

- Smart announcement system for a retirement home (Flask + Vue.js, Raspberry Pi devices).

EDUCATION

Czech Technical University in Prague **2022 – 2027 (expected)**

Faculty of Electrical Engineering, Cybernetics and Robotics (Bc., MSc) *Prague*

Extra courses: functional analysis, 4 semesters of PhD theoretical physics, RTOS, advanced control.
Thesis (master's, in progress, supervisor Ondřej Bajgar): based on the Bayesian IRL research above.
Thesis (bachelor's, supervisor Ing. Jiří Zemánek, Ph.D.): Magnetohydrodynamic (MHD) cooling of a conductive liquid.

- Custom MHD solver in OpenFOAM, real-time MPC with surrogate models, and verification hardware.

COMPETITIONS

- **1st, Vrgineers Algorithm Design Competition** (international, ~25 entrants): real-time stereo 4K image processing in CUDA.
- **1st, Microsoft AI Accessibility Hackathon** (~15 teams): AiAiEyes, an on-device visual aid for the visually impaired, later developed with SONS.
- **1st, Porsche Engineering Student Competition** (national): vision-to-control racing agent with the fastest lap in CARLA.
- **1st, LIFMAT AI healthcare hackathon** (20 teams): digitising and summarising paper medical reports.
- **1st, EuroTeQaThon national round** (15 teams): audio-embedding anomaly detection for brewery machinery.

EXTRACURRICULAR

TePrVe (Technical evening of presentations, teprve.org, press release, Czech only)

- Founded a student meetup (~50 regular attendees) where students give each other talks on their technical projects.

Rakathon 2026 (rakathon.cz) *Main Organiser, Prague*

- National AI-against-cancer hackathon in Prague, Brno and Ostrava: 500+ participants, 24 challenges from hospital oncology practice.

Talks

- **How to backpropagate through fluids**: (TePrVe čtvrtý: video, in Czech) a talk on differentiable fluid sims in NVIDIA Warp
- **CUDY z nudy:** (TePrVe první: link, in Czech) introductory talk on CUDA, but the pun doesn't work in English
- **How to get study notes from slides?** (March 2026) a project for using an agentic LLM harness to convert sparse slides into fluid and well readable notes.
- **LectureRAG:** (AI Tinkerers, link): a talk on creating a knowledge base from lecture recordings
- **Pinakes:** (AI Tinkerers, link): an experimental tool for evaluating how well agents keep notes organised

SKILLS & LANGUAGES

- Python, PyTorch, C++/CUDA, NVIDIA Warp, MuJoCo, ROS2, Linux — Czech, English (fluent)